

Regression tutorial

C5025HF: Step-by-step practical guide

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Setup

We'll keep this simple. Just three essential packages:

```
library(ggplot2)      # for pretty plots
library(dplyr)        # for data wrangling
library(palmerpenguins) # our dataset

# Load the data and remove rows with missing values
data(penguins)
penguins <- penguins %>%
  filter(!is.na(bill_length_mm),
         !is.na(body_mass_g),
         !is.na(species),
         !is.na(flipper_length_mm))

# Take a quick look
head(penguins)
```

```
# A tibble: 6 x 8
  species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
  <fct>   <fct>         <dbl>         <dbl>         <int>         <int>
1 Adelie Torgersen      39.1           18.7           181          3750
2 Adelie Torgersen      39.5           17.4           186          3800
3 Adelie Torgersen      40.3            18           195          3250
4 Adelie Torgersen      36.7           19.3           193          3450
5 Adelie Torgersen      39.3           20.6           190          3650
6 Adelie Torgersen      38.9           17.8           181          3625
# i 2 more variables: sex <fct>, year <int>
```

What we're looking at: Data on 333 penguins from three species. We'll use their physical measurements to understand regression.

Part 1: Simple Linear Regression

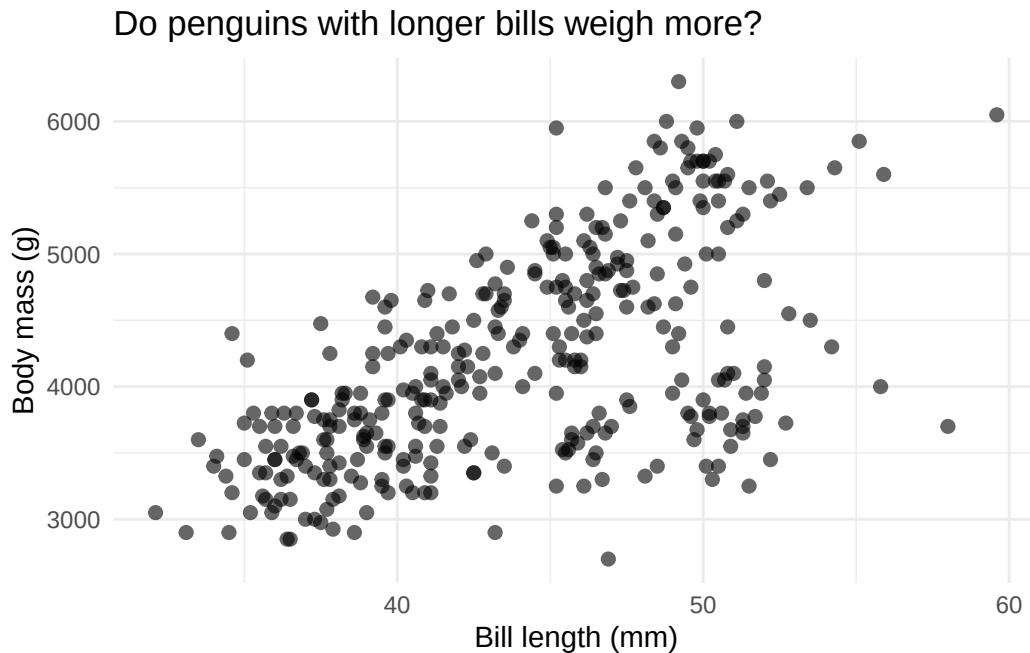
The Question: Does bill length predict body mass?

Let's start with the most basic regression: one predictor, one outcome.

Step 1: Always plot your data first

Never fit a model before looking at your data. Seriously, never.

```
# Create a scatter plot
ggplot(penguins, aes(x = bill_length_mm, y = body_mass_g)) +
  geom_point(alpha = 0.6, size = 2) +
  labs(
    x = "Bill length (mm)",
    y = "Body mass (g)",
    title = "Do penguins with longer bills weigh more?"
  ) +
  theme_minimal()
```



What do we see? There's definitely a positive relationship—longer bills tend to go with heavier penguins. But there's a lot of scatter. Let's quantify this relationship with regression.

Step 2: Fit the model

The `lm()` function fits a linear **m**odel. The formula `y ~ x` means “y is predicted by x”.

```
# Fit the regression: body mass predicted by bill length
model1 <- lm(body_mass_g ~ bill_length_mm, data = penguins)

# The model object is now stored in model1
# Let's see what's inside
class(model1)
```

```
[1] "lm"
```

```
names(model1)
```

```
[1] "coefficients" "residuals"    "effects"      "rank"
[5] "fitted.values" "assign"       "qr"           "df.residual"
[9] "xlevels"      "call"        "terms"        "model"
```

What just happened? We created a model object called `model1`. It's a list containing everything R calculated: coefficients, residuals, fitted values, and more.

Step 3: Look at the output

```
# The summary() function shows us the key results
summary(model1)
```

Call:

```
lm(formula = body_mass_g ~ bill_length_mm, data = penguins)
```

Residuals:

Min	1Q	Median	3Q	Max
-1762.08	-446.98	32.59	462.31	1636.86

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	362.307	283.345	1.279	0.202
bill_length_mm	87.415	6.402	13.654	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 645.4 on 340 degrees of freedom

Multiple R-squared: 0.3542, Adjusted R-squared: 0.3523

F-statistic: 186.4 on 1 and 340 DF, p-value: < 2.2e-16

Let's break this down:

- **Coefficients table:** Shows our intercept and slope
- **Residual standard error:** How far off our predictions typically are (645.43g)
- **R-squared:** Proportion of variance explained ($0.35 = 35\%$)
- **F-statistic:** Tests if the model is better than just using the mean (it is!)

Step 4: Extract and interpret the coefficients

Instead of just reading the summary, let's pull out the numbers we need:

```
# Get the coefficients
coefficients(model1)
```

```
(Intercept) bill_length_mm
362.30672      87.41528
```

```
# Or access them directly from the model object
model1$coefficients
```

```
(Intercept) bill_length_mm
362.30672      87.41528
```

What do these mean?

- **Intercept = 362.31:** A penguin with 0mm bill length would weigh 362.31g. (Obviously impossible—this is just where the line crosses the y-axis)
- **Slope = 87.42:** For every 1mm increase in bill length, body mass increases by **87.42 grams** on average

Let's get confidence intervals too:

```
# 95% confidence intervals for our coefficients
ci1 <- confint(model1)
ci1
```

```
              2.5 %    97.5 %
(Intercept) -195.02364 919.6371
bill_length_mm  74.82279 100.0078
```

We're 95% confident the true slope is between 74.82 and 100.01 grams per mm.

Step 5: Get R-squared

How much of the variation in body mass does bill length explain?

```
# Extract R-squared from the summary
r2_model1 <- summary(model1)$r.squared
r2_model1
```

```
[1] 0.3541557
```

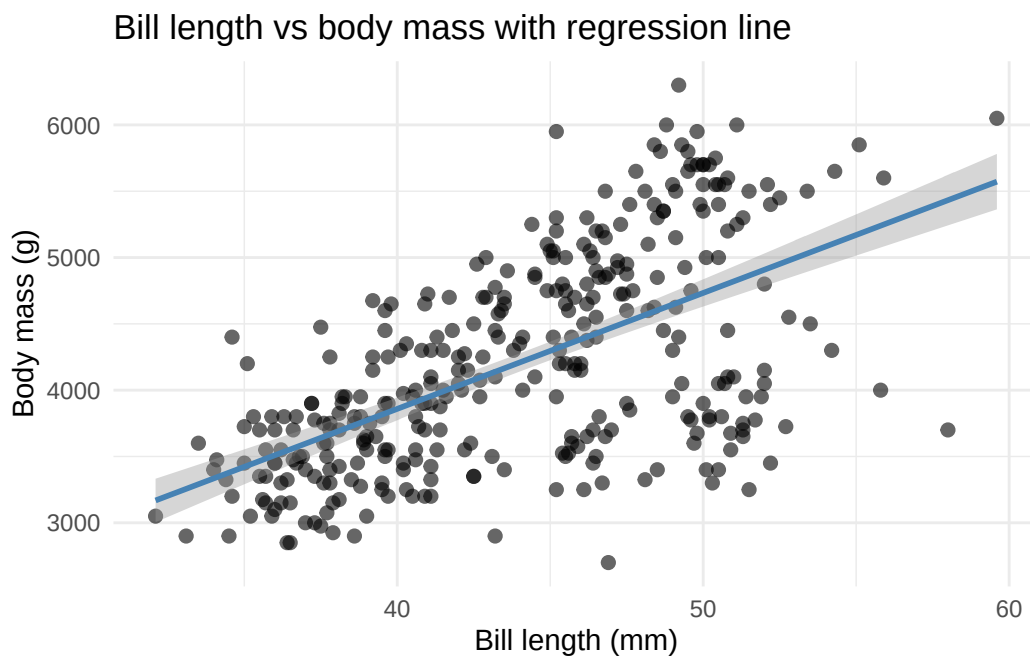
```
# For adjusted R-squared
summary(model1)$adj.r.squared
```

```
[1] 0.3522562
```

35% explained. That means bill length accounts for about a third of why some penguins are heavier than others. Not bad, but there's clearly more to the story.

Step 6: Add the regression line to our plot

```
ggplot(penguins, aes(x = bill_length_mm, y = body_mass_g)) +
  geom_point(alpha = 0.6, size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "steelblue") +
  labs(
    x = "Bill length (mm)",
    y = "Body mass (g)",
    title = "Bill length vs body mass with regression line"
  ) +
  theme_minimal()
```



The shaded area shows our uncertainty about where the true line is.

Step 7: Understand predictions and residuals

Let's manually see what the model predicts:

```
# Get fitted (predicted) values for each penguin
fitted_values <- fitted(model1)

# Get residuals (observed - predicted)
residuals_values <- residuals(model1)

# Look at first 10 penguins
data.frame(
  observed = penguins$body_mass_g[1:10],
  predicted = fitted_values[1:10],
  residual = residuals_values[1:10]
)
```

	observed	predicted	residual
1	3750	3780.244	-30.24405
2	3800	3815.210	-15.21017
3	3250	3885.142	-635.14239
4	3450	3570.447	-120.44739
5	3650	3797.727	-147.72711
6	3625	3762.761	-137.76100
7	4675	3788.986	886.01442
8	3475	3343.168	131.83233
9	4250	4033.748	216.25164
10	3300	3666.604	-366.60419

What's a residual? It's how wrong we were. If a penguin weighs 3750g but we predicted 3600g, the residual is +150g. Positive residuals mean we underestimated; negative means we overestimated.

Part 2: Multiple Regression

The Question: Does adding flipper length improve our predictions?

We saw that bill length explains 35% of body mass variation. Can we do better by adding another predictor?

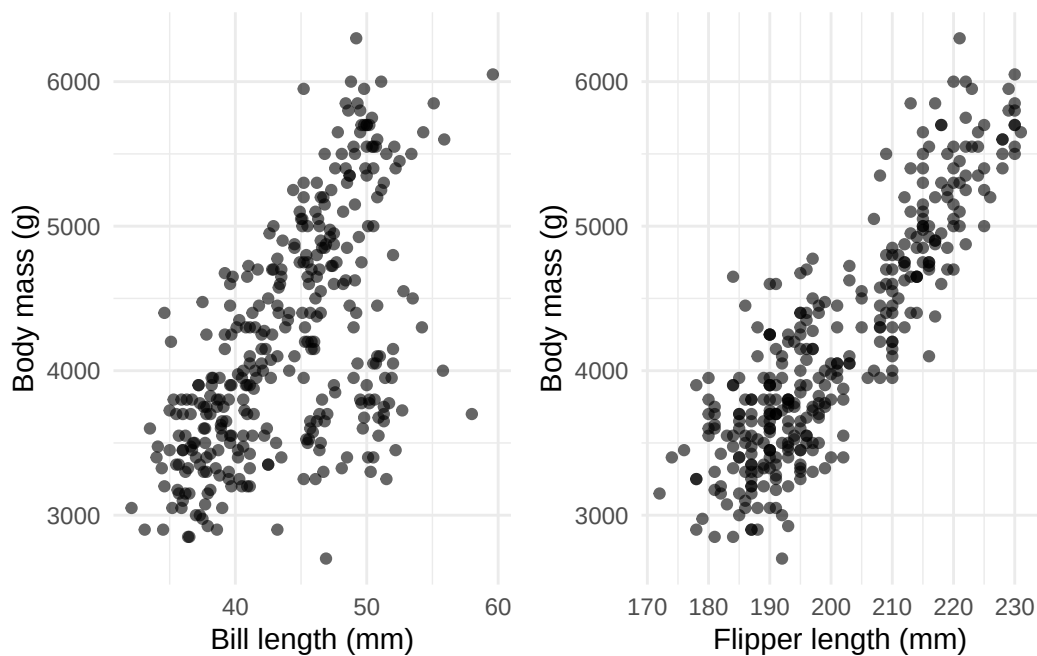
Step 1: Visualize both predictors

```
# Create two plots side by side
library(gridExtra)

p1 <- ggplot(penguins, aes(x = bill_length_mm, y = body_mass_g)) +
  geom_point(alpha = 0.6) +
  labs(x = "Bill length (mm)", y = "Body mass (g)") +
  theme_minimal()

p2 <- ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g)) +
  geom_point(alpha = 0.6) +
  labs(x = "Flipper length (mm)", y = "Body mass (g)") +
  theme_minimal()

grid.arrange(p1, p2, ncol = 2)
```



Both show positive relationships. Now let's include both in one model.

Step 2: Fit the multiple regression

The + sign means “include both predictors”:

```
# Fit model with two predictors
model2 <- lm(body_mass_g ~ bill_length_mm + flipper_length_mm, data = penguins)

summary(model2)
```

Call:

```
lm(formula = body_mass_g ~ bill_length_mm + flipper_length_mm,
    data = penguins)
```

Residuals:

Min	1Q	Median	3Q	Max
-1090.5	-285.7	-32.1	244.2	1287.5

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5736.897	307.959	-18.629	<2e-16 ***
bill_length_mm	6.047	5.180	1.168	0.244
flipper_length_mm	48.145	2.011	23.939	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 394.1 on 339 degrees of freedom

Multiple R-squared: 0.76, Adjusted R-squared: 0.7585

F-statistic: 536.6 on 2 and 339 DF, p-value: < 2.2e-16

Step 3: Interpret the coefficients carefully

This is where multiple regression gets interesting:

```
coefficients(model2)
```

(Intercept)	bill_length_mm	flipper_length_mm
-5736.897161	6.047488	48.144859

```
ci2 <- confint(model2)
ci2
```

	2.5 %	97.5 %
(Intercept)	-6342.64860	-5131.14573

```
bill_length_mm      -4.14117    16.23615
flipper_length_mm   44.18902    52.10069
```

What do these mean?

- **bill_length_mm = 6.05:** Holding flipper length constant, a 1mm increase in bill length increases body mass by 6.05g
- **flipper_length_mm = 48.14:** Holding bill length constant, a 1mm increase in flipper length increases body mass by 48.14g

Wait, what happened? The bill length effect dropped from 87.42g to 6.05g! Why? Because bill length and flipper length are correlated—they both measure “penguin size”. Once we account for flipper length, bill length doesn’t add as much.

Step 4: Compare model fit

Did adding flipper length help?

```
# Model 1 R-squared
r2_model1
```

```
[1] 0.3541557
```

```
# Model 2 R-squared
r2_model2 <- summary(model2)$r.squared
r2_model2
```

```
[1] 0.7599577
```

Huge improvement! We went from 35% to 76% variance explained. Flipper length is clearly an important predictor.

Step 5: Formal model comparison

We can test if model2 is significantly better than model1:

```
# ANOVA compares nested models
anova(model1, model2)
```

Analysis of Variance Table

```
Model 1: body_mass_g ~ bill_length_mm
Model 2: body_mass_g ~ bill_length_mm + flipper_length_mm
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     340 141638626
2     339  52643125   1  88995501 573.09 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Reading the output: The p-value is tiny (< 0.001), confirming that adding flipper length significantly improves the model.

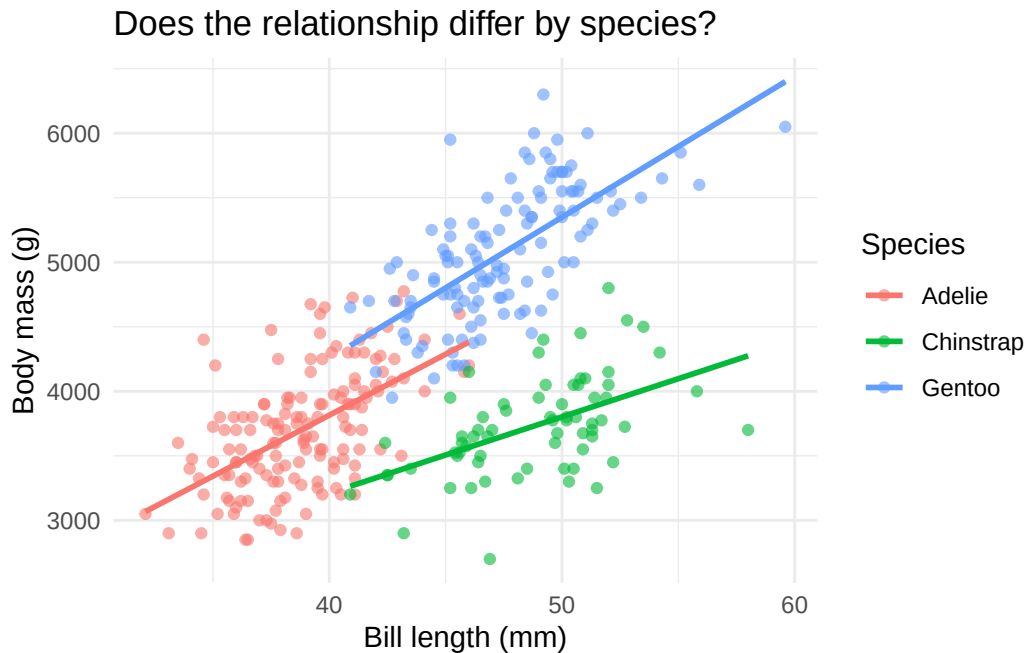
Part 3: Interactions

The Question: Does the bill-length effect differ by species?

Maybe the relationship between bill length and body mass isn't the same for all three species.

Step 1: Plot by species

```
ggplot(penguins, aes(x = bill_length_mm, y = body_mass_g, color = species)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "Bill length (mm)",
    y = "Body mass (g)",
    title = "Does the relationship differ by species?",
    color = "Species"
  ) +
  theme_minimal()
```



What do we see? Three different regression lines with different slopes. Adelie (pink) has a gentler slope than Gentoo (green). This suggests an **interaction**.

Step 2: Fit the interaction model

The * symbol means “include both main effects AND their interaction”:

```
# Main effects + interaction
model3 <- lm(body_mass_g ~ bill_length_mm * species, data = penguins)

summary(model3)
```

Call:

```
lm(formula = body_mass_g ~ bill_length_mm * species, data = penguins)
```

Residuals:

Min	1Q	Median	3Q	Max
-918.76	-245.89	-8.65	238.44	1126.27

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
----------	------------	---------	----------

```

(Intercept)                34.88      443.18    0.079    0.937
bill_length_mm             94.50       11.40    8.291 2.73e-15 ***
speciesChinstrap          811.26      799.81    1.014    0.311
speciesGentoo            -158.71      683.19   -0.232    0.816
bill_length_mm:speciesChinstrap -35.38      17.75   -1.994    0.047 *
bill_length_mm:speciesGentoo   14.96      15.79    0.948    0.344
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 371.8 on 336 degrees of freedom
Multiple R-squared:  0.7882,    Adjusted R-squared:  0.7851
F-statistic: 250.1 on 5 and 336 DF,  p-value: < 2.2e-16

```

Step 3: Interpret interaction coefficients

This is the trickiest part of regression. Let's go slowly:

```
coefficients(model3)
```

```

              (Intercept)                bill_length_mm
              34.88299                94.49982
speciesChinstrap                speciesGentoo
              811.26034                -158.71092
bill_length_mm:speciesChinstrap  bill_length_mm:speciesGentoo
              -35.38208                14.95935

```

How to read this:

1. **Intercept** and **bill_length_mm**: These apply to the *reference species* (Adelie, alphabetically first)
2. **speciesChinstrap** and **speciesGentoo**: These are *adjustments* to the intercept for those species
3. **bill_length_mm:speciesChinstrap** and **bill_length_mm:speciesGentoo**: These are *adjustments* to the slope for those species

Species-specific slopes:

```

# Extract coefficients
coef3 <- coefficients(model3)

# Calculate species-specific slopes

```

```

slope_adelie <- coef3["bill_length_mm"]
slope_chinstrap <- coef3["bill_length_mm"] + coef3["bill_length_mm:speciesChinstrap"]
slope_gentoo <- coef3["bill_length_mm"] + coef3["bill_length_mm:speciesGentoo"]

# Display them
data.frame(
  Species = c("Adelie", "Chinstrap", "Gentoo"),
  Slope = round(c(slope_adelie, slope_chinstrap, slope_gentoo), 2)
)

```

	Species	Slope
1	Adelie	94.50
2	Chinstrap	59.12
3	Gentoo	109.46

- Adelie: 94.5 g/mm
- Chinstrap: 59.12 g/mm
- Gentoo: 109.46 g/mm

Step 4: Do we need the interaction?

Test if the interaction terms improve the model:

```

# Model without interaction
model3_simple <- lm(body_mass_g ~ bill_length_mm + species, data = penguins)

# Compare
anova_comparison <- anova(model3_simple, model3)
anova_comparison

```

Analysis of Variance Table

Model 1: body_mass_g ~ bill_length_mm + species

Model 2: body_mass_g ~ bill_length_mm * species

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	338	47613707				
2	336	46447006	2	1166702	4.22	0.01549 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The p-value is 0.015. Since this is less than 0.05, the interaction **is** statistically significant. The slopes really do differ across species. We should keep the interaction in the model!

Part 4: ANOVA is Just Regression

The Big Reveal: ANOVA = Regression with categorical predictors

This blows people's minds every time. Let's prove it.

Traditional ANOVA approach

Do the three species have different average body masses?

```
# Fit ANOVA
aov_model <- aov(body_mass_g ~ species, data = penguins)
aov_summary <- summary(aov_model)
aov_summary
```

```
              Df    Sum Sq Mean Sq F value Pr(>F)
species        2 146864214 73432107   343.6 <2e-16 ***
Residuals     339  72443483   213698
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANOVA says: $F = 343.63$, $p < 0.001$. Species significantly affect body mass.

Regression approach

Now let's use `lm()` with a categorical predictor:

```
# Fit regression with species as predictor
lm_model <- lm(body_mass_g ~ species, data = penguins)
lm_summary <- summary(lm_model)
lm_summary
```

Call:

```
lm(formula = body_mass_g ~ species, data = penguins)
```

Residuals:

Min	1Q	Median	3Q	Max
-1126.02	-333.09	-33.09	316.91	1223.98

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3700.66	37.62	98.37	<2e-16 ***
speciesChinstrap	32.43	67.51	0.48	0.631
speciesGentoo	1375.35	56.15	24.50	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 462.3 on 339 degrees of freedom

Multiple R-squared: 0.6697, Adjusted R-squared: 0.6677

F-statistic: 343.6 on 2 and 339 DF, p-value: < 2.2e-16

Notice the bottom line: **F-statistic: 343.63, $p < 0.001$**

They're identical!

```
# Get ANOVA table from the lm model
anova(lm_model)
```

Analysis of Variance Table

Response: body_mass_g

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	146864214	73432107	343.63	< 2.2e-16 ***
Residuals	339	72443483	213698		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Same F-statistic, same p-value, same conclusion. ANOVA is just regression where the predictor is categorical instead of continuous.

What do the regression coefficients mean?

```
coef_lm <- coefficients(lm_model)
coef_lm
```

(Intercept)	speciesChinstrap	speciesGentoo
3700.66225	32.42598	1375.35401

- **Intercept (3700.66)**: Mean body mass for Adelie (the reference group)
- **speciesChinstrap (32.43)**: Chinstrap penguins weigh 32.43g less than Adelie, on average
- **speciesGentoo (1375.35)**: Gentoo penguins weigh 1375.35g more than Adelie, on average

Let's verify this by calculating group means directly:

```
# Calculate means by species
species_means <- penguins %>%
  group_by(species) %>%
  summarise(mean_mass = mean(body_mass_g))
species_means
```

```
# A tibble: 3 x 2
  species    mean_mass
  <fct>      <dbl>
1 Adelie    3701.
2 Chinstrap 3733.
3 Gentoo    5076.
```

- Adelie: 3700.66g (= intercept)
- Chinstrap: 3733.09g = 3700.66 + 32.43
- Gentoo: 5076.02g = 3700.66 + 1375.35

The regression coefficients directly tell us group differences!

Part 5: Checking Your Model (Diagnostics)

Why diagnostics matter

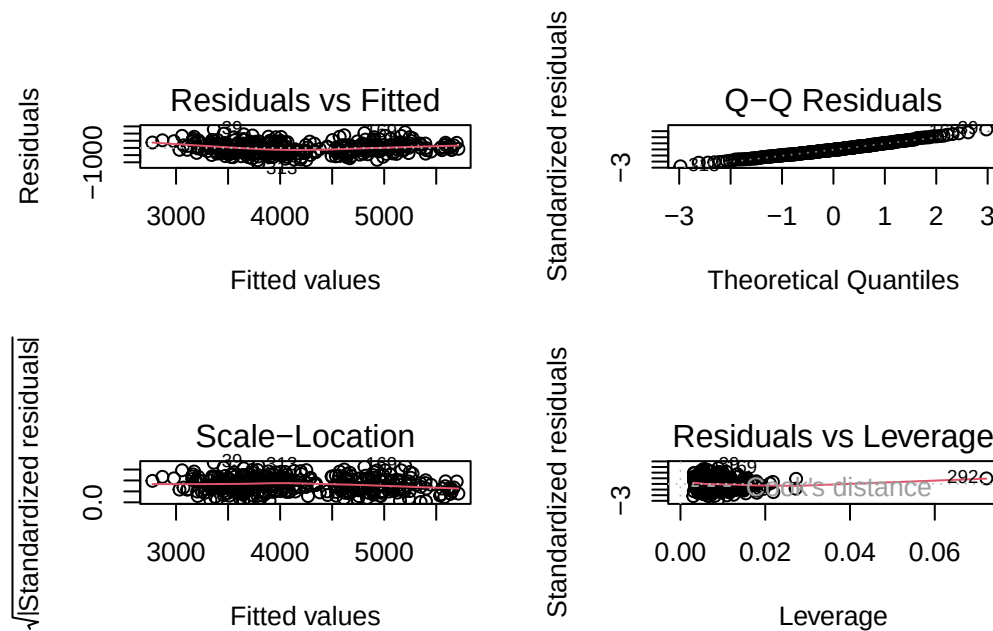
Regression makes assumptions. If those assumptions are violated, your p-values and confidence intervals are unreliable. We need to check:

1. **Linearity:** Is the relationship actually linear?
2. **Homoscedasticity:** Is the variance constant?
3. **Normality:** Are residuals roughly normal?
4. **Independence:** Are observations independent? (Check by study design)

The four diagnostic plots

R gives us four helpful plots. Let's use model2 (body mass ~ bill length + flipper length):

```
# Create all four diagnostic plots
par(mfrow = c(2, 2)) # 2x2 grid of plots
plot(model2)
```



```
par(mfrow = c(1, 1)) # reset to single plot
```

How to read each plot

Plot 1: Residuals vs Fitted

- **What to look for:** Random scatter around the horizontal line at zero
- **Red flag:** A pattern (curve, U-shape) means the relationship isn't linear
- **Red flag:** A funnel shape (wider on one side) means variance isn't constant

Plot 2: Normal Q-Q

- **What to look for:** Points following the diagonal line
- **Red flag:** Points curving away from the line means residuals aren't normal
- **Note:** Small deviations at the ends are usually okay

Plot 3: Scale-Location

- **What to look for:** Random scatter, roughly flat red line
- **Red flag:** Red line sloping up means variance increases with fitted values
- **This checks:** Homoscedasticity (constant variance)

Plot 4: Residuals vs Leverage

- **What to look for:** All points inside the dashed lines (Cook's distance)
- **Red flag:** Points outside the dashed lines have high influence
- **What it means:** These observations strongly affect the regression line

What does model2 look like?

Looking at our four plots:

- ☒ **Residuals vs Fitted:** Pretty random, no strong pattern
- ☒ **Q-Q plot:** Points mostly follow the line; residuals are roughly normal
- ☒ **Scale-Location:** Fairly flat; variance looks constant
- ☒ **Residuals vs Leverage:** No points way outside; no highly influential observations

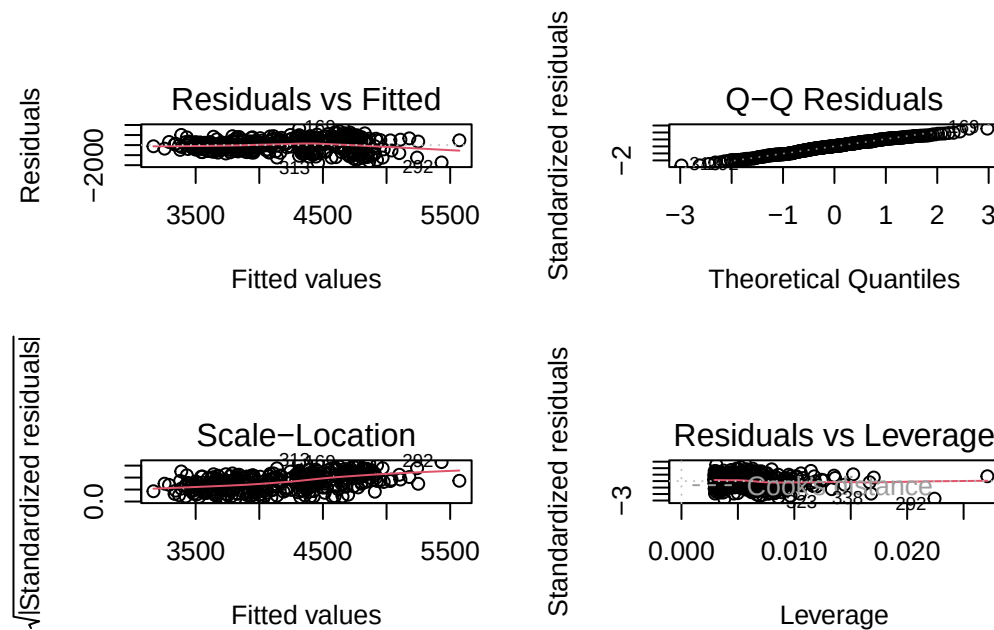
Verdict: This model looks good! Assumptions are reasonably met.

When things go wrong: An example

Let's see what bad diagnostics look like. We'll fit a model that doesn't account for species:

```
# Deliberately ignoring species
bad_model <- lm(body_mass_g ~ bill_length_mm, data = penguins)

par(mfrow = c(2, 2))
plot(bad_model)
```



```
par(mfrow = c(1, 1))
```

What's wrong here?

- **Residuals vs Fitted:** You can see some pattern/grouping—we're missing an important predictor (species)
- **Q-Q plot:** Tails deviate more from the line
- This shows why model2 (which includes flipper length) is better

Summary: The Essential R Commands

Here's everything you need to know:

```
# Fit a model
model <- lm(y ~ x, data = mydata)           # simple regression
model <- lm(y ~ x1 + x2, data = mydata)     # multiple regression
model <- lm(y ~ x1 * x2, data = mydata)     # with interaction

# Look at results
summary(model)                             # full output
coefficients(model)                        # just the coefficients
confint(model)                             # confidence intervals
anova(model)                               # ANOVA table

# Extract components
model$coefficients                         # coefficients
fitted(model)                             # predicted values
residuals(model)                          # residuals
summary(model)$r.squared                   # R-squared

# Compare models
anova(model1, model2)                      # test if model2 is better

# Diagnostics
plot(model)                                # creates 4 diagnostic plots
```

Practice Exercises

Try these yourself:

1. Fit a simple regression predicting `body_mass_g` from `flipper_length_mm`. What's the slope? Interpret it in plain English.
2. Add `bill_length_mm` to create a multiple regression. How did the flipper length coefficient change?
3. Fit separate regressions for each species (use `filter()`). Do Adelie, Chinstrap, and Gentoo have different slopes?

4. Use `lm()` to test if body mass differs between male and female penguins. Then verify your answer using `t.test()`.
 5. Check the diagnostics for your models. Do any assumptions look violated?
-

What You've Learned

- How to fit linear models with `lm()`
- How to extract and interpret coefficients
- The difference between simple and multiple regression
- What interactions mean and how to test them
- That ANOVA is secretly just regression
- How to check if your model assumptions are met

Most importantly: You learned to interrogate model objects directly using base R, not relying on fancy packages. This is how you really understand what's going on under the hood.

Now go practice with your own data!